

Combinatorial Theory: Positroid / 9.14.26

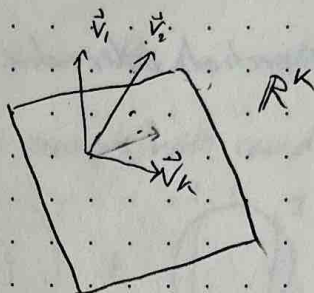
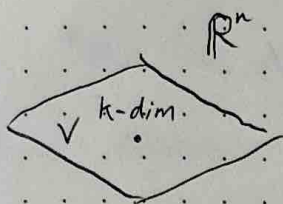
Let $\mathbb{F}$ be a field (say $\mathbb{F} = \mathbb{R}$), then

$V =$ row space of A
 $\vec{v}_1, \dots, \vec{v}_n$ are the columns of A

$$Gr_{kn} = \{k\text{-dim subspaces in } \mathbb{F}^n\}$$

$$= \{ \text{ ~~} k \times k \text{ matrices of rank } k \text{ } \} / GL_k~~$$

$$= \{ \text{configurations of vectors } \vec{v}_1, \dots, \vec{v}_n \text{ span: } \mathbb{F}^k \} \text{ mod } GL_k \text{ action}$$



We also have

$$Gr_{kn} = \sqcup \text{ Schubert cells}$$

recall that $X_\lambda^0 = X_I^0 = X_d^0$

→ Young Diagrams: $\lambda \leq k \times (n-k)$ (shape of $\hat{RREF}$ of A)

→ k -subsets $I \in \binom{[n]}{k}$ (pivot columns in $RREF$ of A)

→ dimension vectors $d = (d_1, \dots, d_n)$ w/ $d_i = \dim(V \cap \langle e_1, \dots, e_{i-1} \rangle)$ such that $0 \leq d_1 \leq \dots \leq d_n$.

EX: let $k=3, n=7$

x	x	x	x	0	0	0
x	x	x	0	0	0	0
x	0	0	0	0	0	0
0	0	0	0	0	0	0

$$\lambda = (3, 3, 1)$$

$$I = \{1, 2, 4, 5\}$$

$$d = (0, 1, 1, 1, 2, 3, 3)$$

ask for d -vector

The Brhat order for Gr_{kn} is the partial order on Schubert cells, (given by containment of their closures)

↳ some may be familiar w/ the permutation ordering.
This is related, but looks different

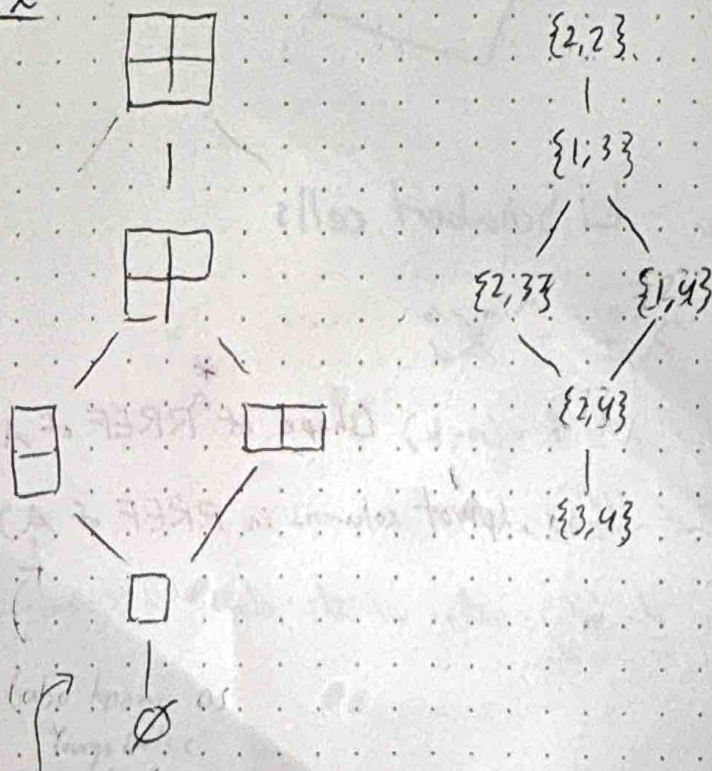
Lemma (TFAE):

A) $\lambda \leq \lambda'$

B) $I \leq I'$ (termwise order $i_1 \leq i'_1, i_2 \leq i'_2$)

C) all termwise partial order $d_1 \leq d'_1, d_2 \leq d'_2, \dots$

EX:



(also known as "Young's lattice for a fixed λ ")

(Recall) Plücker coords on $Gr_{k,n}$.

$$\Delta_I = \text{max. minor of } A \text{ in column set } I$$

where $I \in \binom{[n]}{k}$ and $Gr_{k,n} \hookrightarrow \mathbb{P}^{\binom{n}{k}-1}$ can be embedded.

From here, we have the highly applicable Plücker Relations

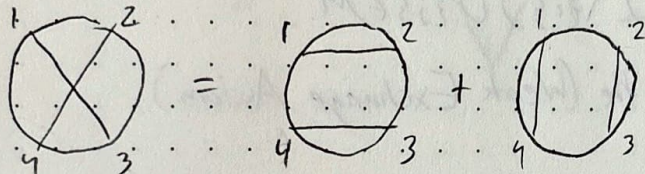
→ (see typed notes for applications)

EX: let $k=2, n=4$. Then $Gr_{2,4} \hookrightarrow \mathbb{P}^{\binom{4}{2}-1}$.

$$\Delta_{13} \cdot \Delta_{24} = \Delta_{12} \cdot \Delta_{34} + \Delta_{14} \cdot \Delta_{23}$$

* this is an example of 3-term Plücker relations.

We can represent this 3-term example with coord diagrams



More generally, take

$$\Delta_{(i_1, \dots, i_k)} = \begin{cases} \text{sgn}(w) \cdot \Delta_{\{w(i_1) < w(i_2) < \dots < w(i_k)\}} & w \in S_n \\ 0 & \text{if there are repeated indices} \end{cases}$$

Def'n (General Plücker Relations):

↳ for all k -tuples $(i_1, \dots, i_k), (j_1, \dots, j_k)$ we have check

$$\Delta_{(i_1, \dots, i_k)} \cdot \Delta_{(j_1, \dots, j_k)} = \sum_{s=1}^n \Delta_{(i_1, \dots, i_k)} \cdot \Delta_{(j_1, \dots, j_{s-1}, i_s, j_{s+1}, \dots, j_k)}$$

Thm: the vanishing ideal of the image of $Gr_{k,n}$ on $\mathbb{P}^{\binom{n}{k}-1}$ is the ideal generated by Plücker relations.

Def'n (Matroids #1):

"ground set"

↳ A nonempty subset $M \subseteq \binom{[n]}{k}$ is a matroid of rank k on n elements. Also, elements $I \in M$ are bases of M , if the following are satisfied

→ for all $I, J \in M$, and for all $i \in I$, there is $j \in J$ such that both

$$(I \setminus \{i\}) \cup \{j\} \text{ and } (J \setminus \{j\}) \cup \{i\}$$

are in M . (Strong Exchange Axiom)

[Def'n (Matroids #2): ← standard def

↳ same as (#1), but using $I, J \in M$ st. for all $i \in I$, there is $j \in J$ st.

$$(I \setminus \{i\}) \cup \{j\} \in M$$

(this is called the Weak Exchange Axiom)

→ (Exercise): show that strong and weak exchange axioms are equivalent.

Hint: induction

we can also define matroids using the "Permuted Bruhat order for G_{rank} "

↳ " $\leq_w$ " for a permutation $w \in S_n$

→ for our purposes, a partial order on $\binom{[n]}{k}$ such that

$$I \leq_w J \text{ if } w(I) \leq w(J)$$

↑
termwise

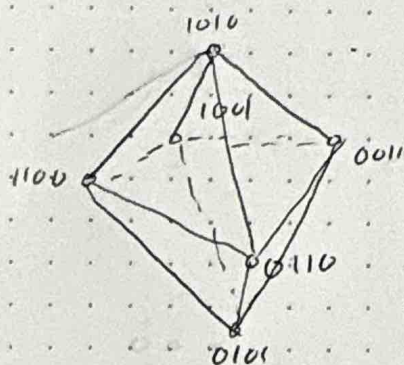
Def'n (Matroid #3):

↳ A nonempty subset $M \subseteq \binom{[n]}{k}$ is a matroid if $\forall w \in S_n$, M has a unique minimal element under $\leq_w$.

→ (Exercise) show def'n (#3) is equivalent to (#2), (#1)

We can also define matroids using polytopes (hypersimplices)

EX: let $k=2$, $n=4$



← odd

